1.33 Differentiation: Graphs of Gradient Functions

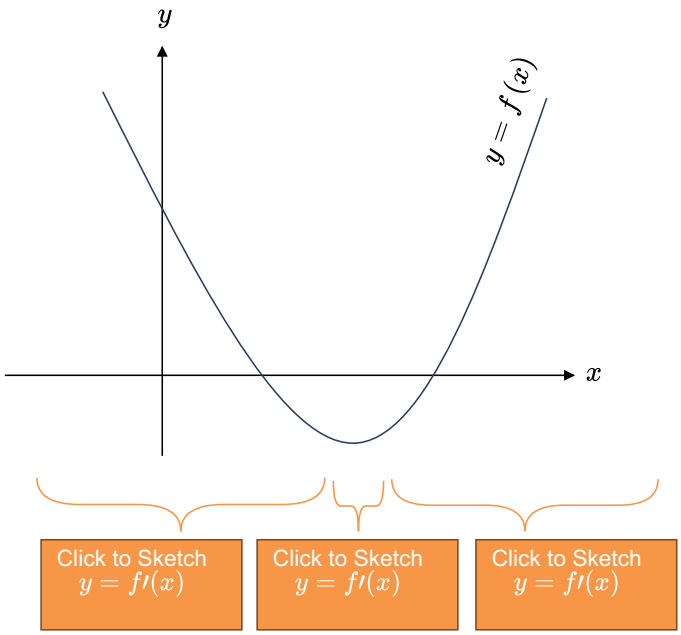
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Sketching Gradient Functions

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The new A Level specification specifically mentions being able to sketch $y = f\prime(x)$.

If you know the function $f\prime(x)$ explicitly (e.g. because you differentiated $y = f(x)$), you can use your knowledge of sketching straight line/ quadratic/cubic graphs. But in other cases **you won't be given the function explicitly**, but **just the sketch**.

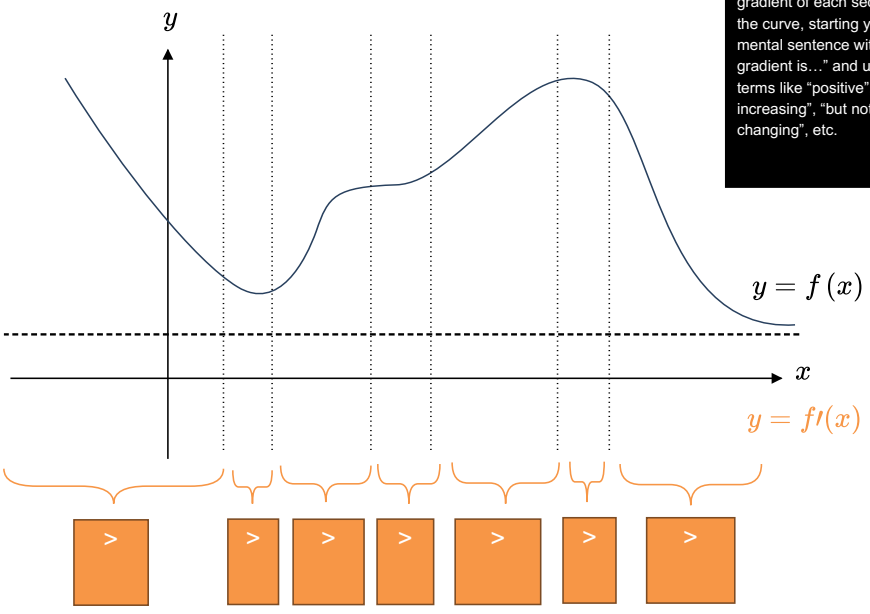


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A Harder One

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Tip: Mentally describe the gradient of each section of the curve, starting your mental sentence with "The gradient is…" and using terms like "positive", "but increasing", "but not changing", etc.

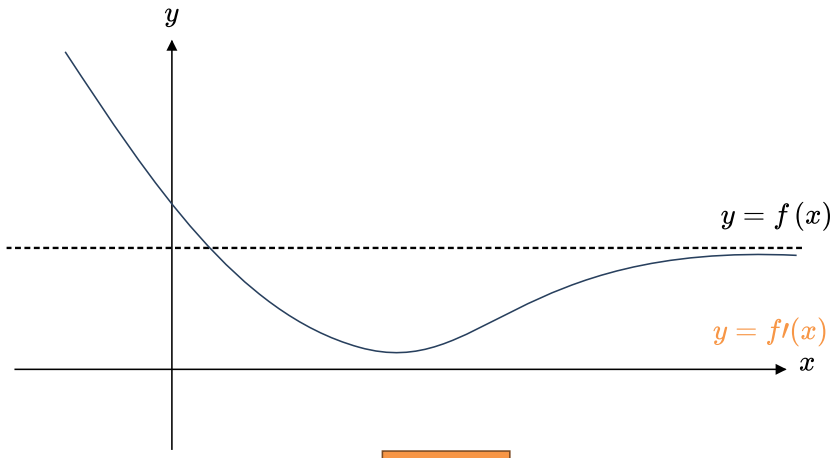


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Test Your Understanding

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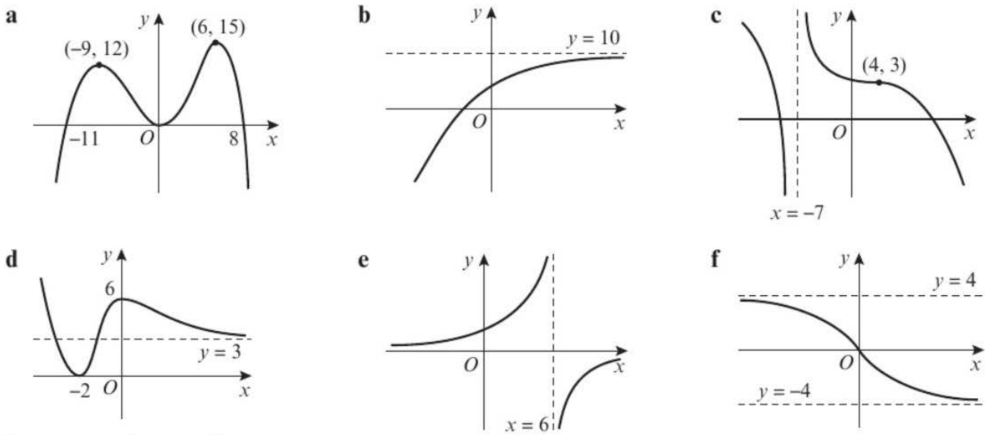
Solution >

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Exercise 12J – Pearson Pure Y1 page 278

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1 For each graph given, sketch the graph of the corresponding gradient function on a separate set of axes. Show the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.



Solutions here

1.33 Differentiation: Graphs of Gradient Functions

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2\quad $\mathrm{f}(x) = (x+1)(x-4)^2$.
a Sketch the graph of $y = \mathrm{f}(x)$.
b On a separate set of axes, sketch the graph of $y = \mathrm{f}'(x)$.
c Show that $\mathrm{f}'(x) = (x-4)(3x-2)$.
d Use the derivative to determine the exact coordinates of the points where the gradient function cuts the coordinate axes.

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Extension

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1 [MAT 2015 1B]

$$f(x) = (x+a)^n$$

where a is a real number and n is a positive whole number, and $n \geq 2$. If $y = f(x)$ and $y = f\prime(x)$ are plotted on the same axes, the number of intersections between $f(x)$ and $f'(x)$ will:

A) always be odd
B) always be even
C) depend on a but not n
D) depend on n but not a
E) depend on both a and n.
F)

Recall that:
Odd function:
$f(-x) = -f(x)$
Even function: $f(-x) = f(x)$